

Avinash.M
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**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

**COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07**

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

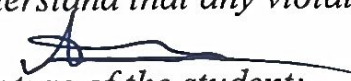
Assessment Tool Weightage: 30%

Software: Cisco Packet Tracer, Wireshark

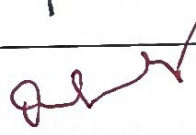
Code of Conduct:

I Avinash.M (192521196) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.


Signature of the student:

Criteria	Weightage	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Problem Identification	20%	5	2	1	1
Debugging	25%	1	1	1	1
Optimization	20%	1	1	1	1
Analysis	20%	1	1	1	1
Code Quality	15%	1	1	1	1

25/30 
Annexure A

Debugging & Optimisation Task

1. A network administrator has configured a workstation with the following settings:

- **IP Address:** 192.168.10.65
- **Subnet Mask:** 255.255.255.192 (/26)
- **Default Gateway:** 192.168.10.1

However, the workstation is unable to communicate with other systems configured in the subnet 192.168.10.0/26.

Task

2. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

3. An organization has allocated the network 192.168.1.0/24 into several departments using the subnet masks /26, /27, and /28. Some departments report insufficient IP addresses, while others have a large number of unused addresses.

Task

Analyze and optimize the subnet allocation by:

- Identifying the network address, broadcast address, and host range for each subnet.
- Calculating the number of usable and unused IP addresses.
- Identifying inefficient subnet allocations.
- Debugging the addressing plan by locating subnet allocation mistakes.
- Recommending an optimized subnetting scheme using VLSM to improve address utilization while supporting current departmental requirements.

4. Two devices are configured with the following IPv6 addresses:

- Device A: 2001:db8::ff00:42:8329/64
- Device B: 2001:db8:0:0:1::1/64

Although both devices are connected to the same network, communication fails.

Task

Debug the IPv6 configuration by:

- Expanding both IPv6 addresses into their complete notation.
- Determining the network prefix of each device.
- Verifying whether both devices belong to the same subnet.
- Identifying the addressing or prefix mismatch.
- Optimizing the IPv6 configuration by assigning appropriate addresses.
- Calculating the number of available host addresses within the corrected subnet.

5. A router contains the following routing table.

Destination Network

100.168.1.0/24

Next Hop

10.0.0.2

Debug the routing configuration by:

- Identifying the matching routing entry using the longest-prefix match.
- Determining the next-hop router selected for packet forwarding.
- Verifying whether the destination IP belongs to the intended subnet.
- Identifying any routing configuration errors.
- Optimizing the routing table to improve packet forwarding efficiency.
- Identifying the route that will be used if no matching destination exists.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Identification	20%	Accurately identifies all errors and root causes with clear understanding	Identifies most errors with minor gaps	Identifies some errors but misses key issues	Unable to identify errors correctly
Debugging	25%	Applies systematic debugging techniques effectively to resolve issues	Uses appropriate debugging methods with minor errors	Limited debugging approach; requires guidance	Unable to debug or resolve issues
Optimization	20%	Optimizes code by significantly improving time complexity using efficient algorithms	Improves time complexity with minor gaps in efficiency	Limited improvement in time complexity	No improvement; inefficient approach retained
Analysis	20%	Demonstrates strong logical reasoning and clear justification of solutions	Shows reasonable analysis with some justification	Limited analysis; weak reasoning	No analytical reasoning demonstrated
Code Quality	15%	Produces clean, well-	Code is mostly	Code lacks structure and	Poorly written code with no

C02 - AT3

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Course : Computer Networks

Code : CSA0707

1. IPV4 Debugging

Given,

* IP Address : 192.168.10.65

* Subnet Mask : 255.255.255.192 (126)

* Default Gateway : 192.168.10.1

Ans :

1) Subnet : 192.168.10.64/26

2) Network Address : 192.168.10.64

3) Broadcast Address : 192.168.10.65 - 192.168.10.126

4) IP Address Valid? = Yes.

5) Gateway valid? No (192.168.10.65) belongs to subnet 192.168.10.0/26 not 192.168.10.64/26

6) Configuration Error : Incorrect default gateway.

8) Configuration Error : Incorrect default gateway

9) Usable Hosts : 62 ($2^6 - 2$)

2) IPv6 Debugging.

Device A : 2001:db8::ff00:42:8329/64

Expanded : 2001:0db8:0000:0000:0000:ff0d+0422:
8329

Device D : 2001:db8:0:0:1:1/64

Expanded : 2001:0db8:0000

Network Prefix:

Device A : 2001:db8:0:0::1/64

Device B : 2001:db8:0:0:/64

Network Prefix:

Device A : 2001:db8:0:0::1/64

Device B : 2001:db8:0:0:/64

Same subnet:

Yes both are in 2001:db8:0:0/64

Issue: Addressing correct : communication failure is likely optimized addresses.

* Device A : 2001:db8:0:0:10/64

* Device B : 2001:db8:0:0:20/64

$2^{64} \approx 18.4$ quintillion addresses.

Subnet Optimization:

Subnet	Network	Host Range
/26	192.168.10	192.168.1.1 - 192.168.162
/27	192.168.164	192.168.1.65 - 192.168.174
/28	192.168.196	192.168.197 - 192.168.198.110

Unused Address.

★) /26 → 62 useable

★) /27 → 30 useable

★) /28 → 14 useable

Inefficient Allocation.

★) Large department should receive /26

★) medium departments should receive /27

★) Small department should receive /28

Optimized VLSM.

★) Largest department → /26

★) medium department → /24

★) Small department → /28

★) Address remaining space to future department

4) Ipv6 Debugging :

Expanded Address A.

2001:0db8:0000:0000:0000:ff00:0042:5327

Expanded Address B.

2001:0db8:0000:0000:0001:0000:0000:0001

Prefix : 2001:db8:00:64

Same Subnet : Yes.

Optimized Address.

A → 2001:db8:0:0::10/64

B → 2001:db8:0:0::20/64

Available Hosts. 2^{64}

5) Routing Table Debugging.

Routing Table.

Destination

Next Hop

192.168.1.0/24

10.0.0.3

192.168.2.0/24

10.0.0.1

Destination : 192.168.2.100

Ans : Matching Route : 192.168.2.0/24

Largest Prefix Match : 192.168.2.0/24

Possible Error : Incorrect or unstable next

Optimization : Verify the route to 10.0.0.3,

Remove Redundant route & main connection
Routing..

If No matching Route Exists : Default Route
10.0.0.1.